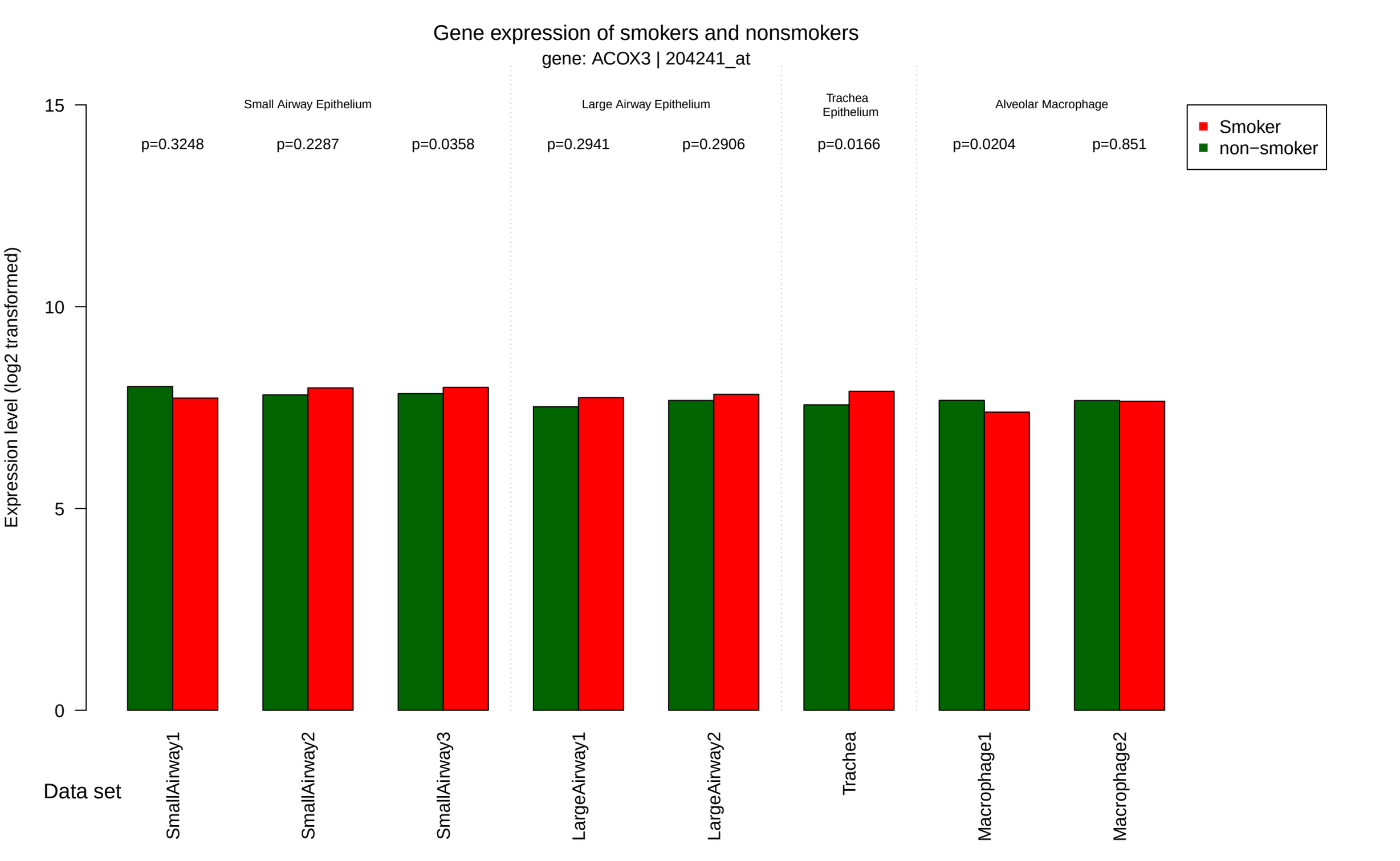
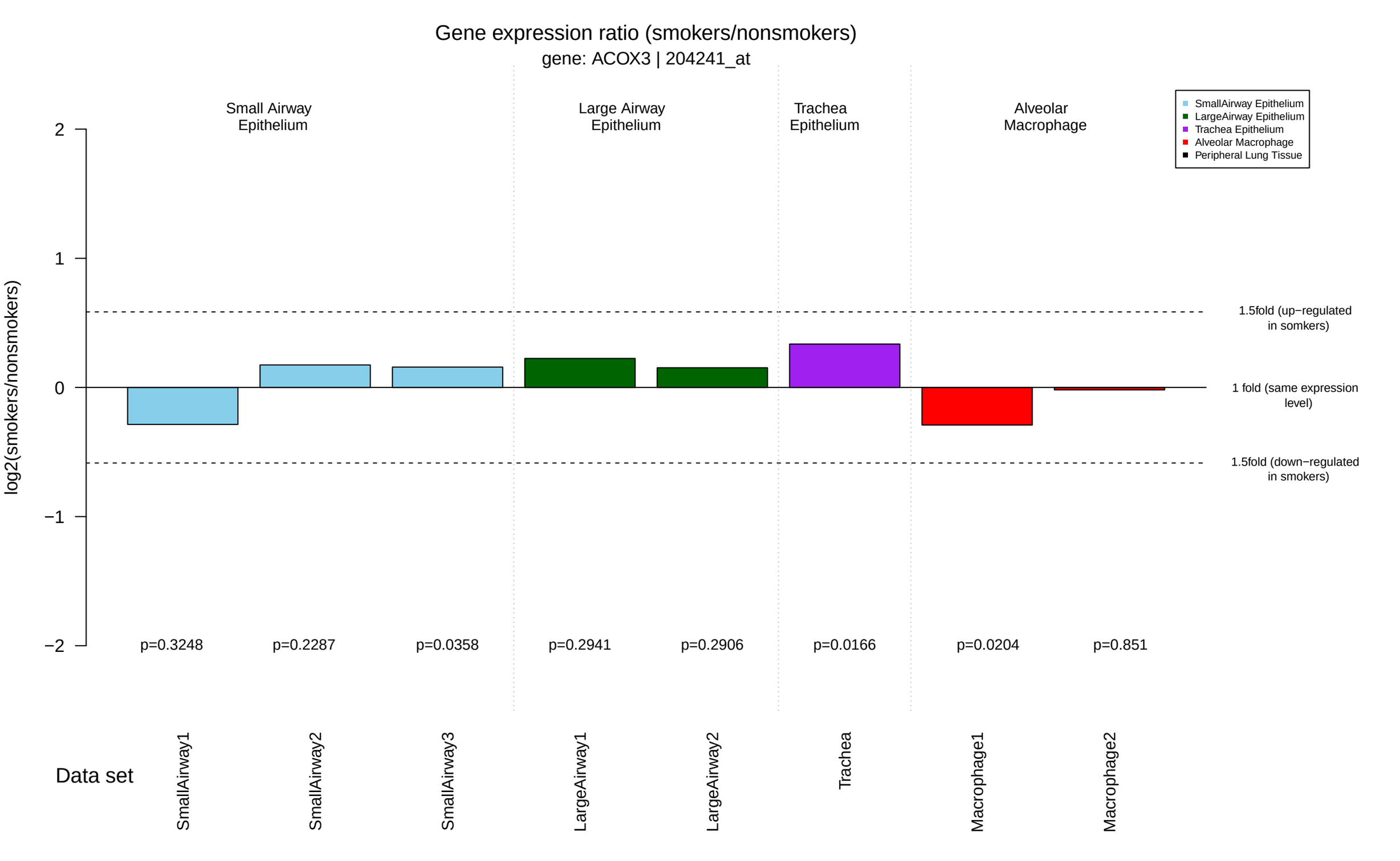
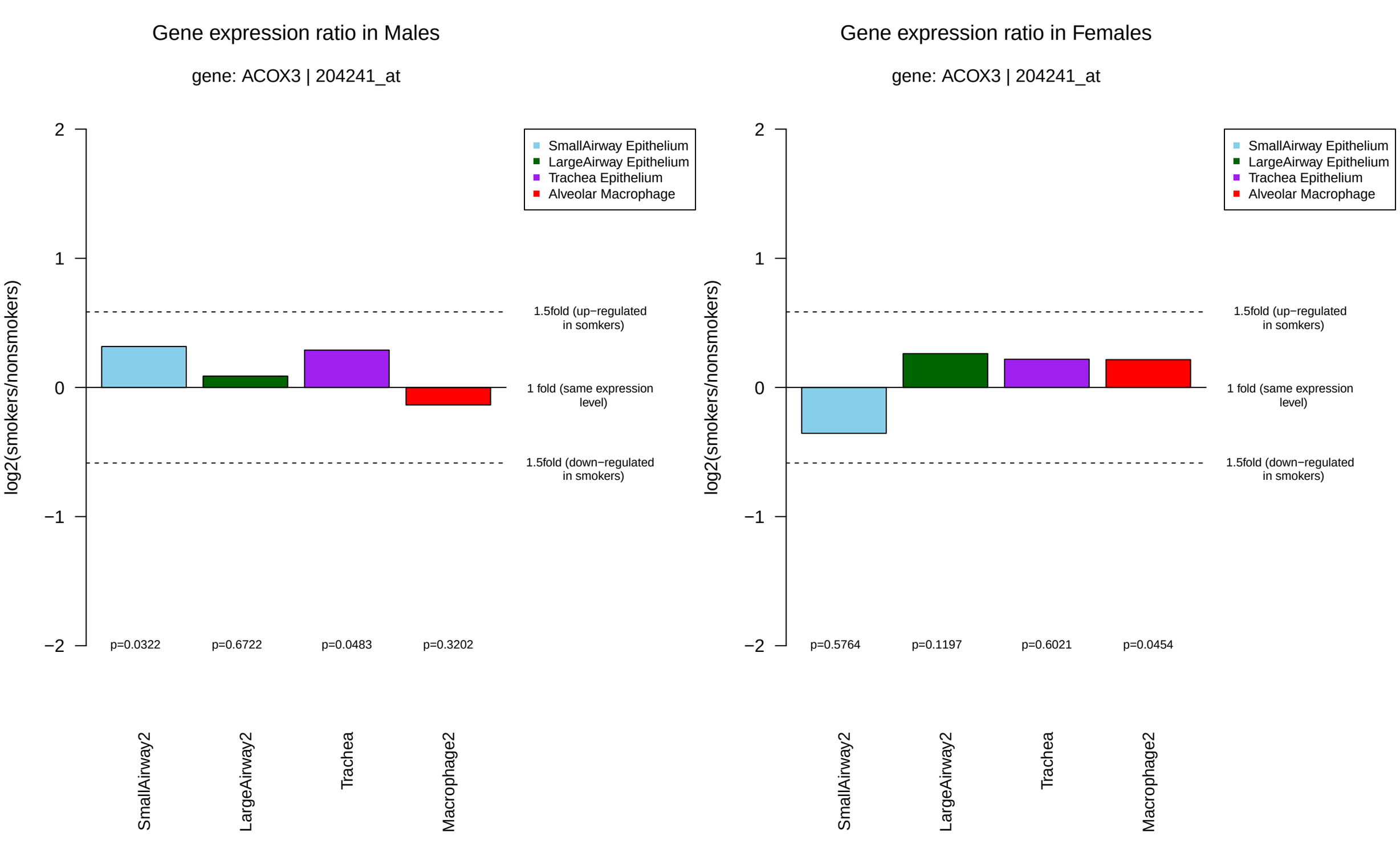
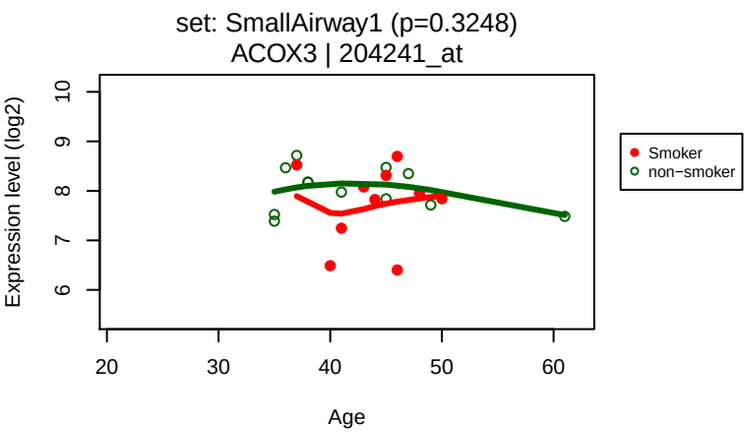
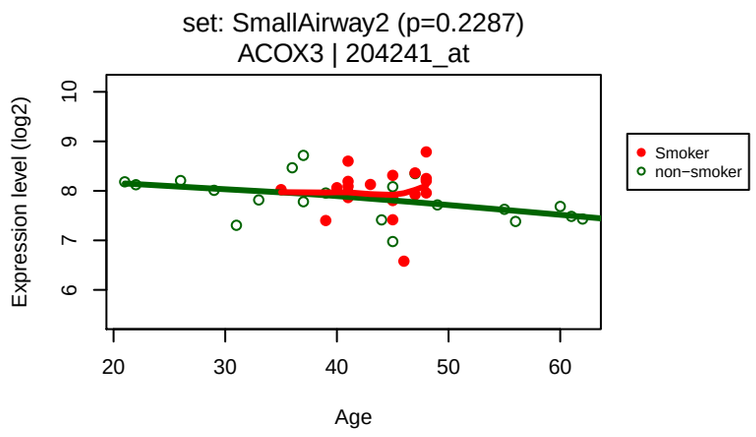
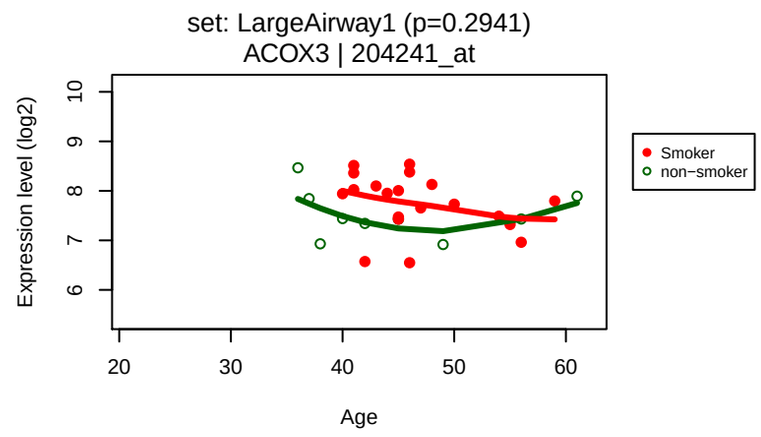

Gene expression of smokers and nonsmokers
gene: ACOX3 | 204241_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3248
p=0.2287
p=0.0358
p=0.2941
p=0.2906
p=0.0166
p=0.0204
p=0.851
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

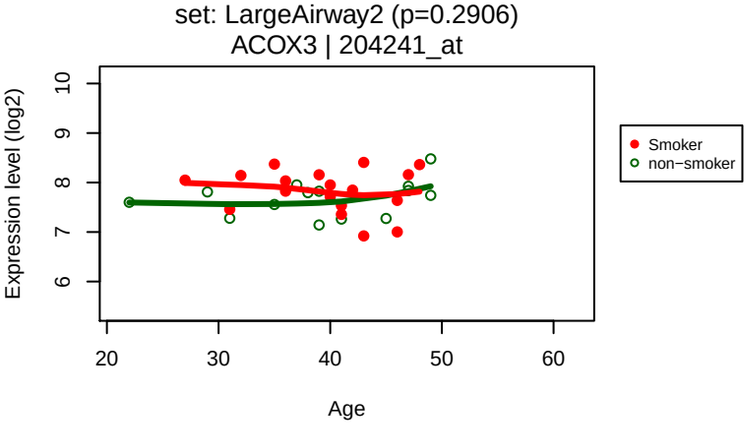

Gene expression ratio (smokers/nonsmokers)
gene: ACOX3 | 204241_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3248
p=0.2287
p=0.0358
p=0.2941
p=0.2906
p=0.0166
p=0.0204
p=0.851
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

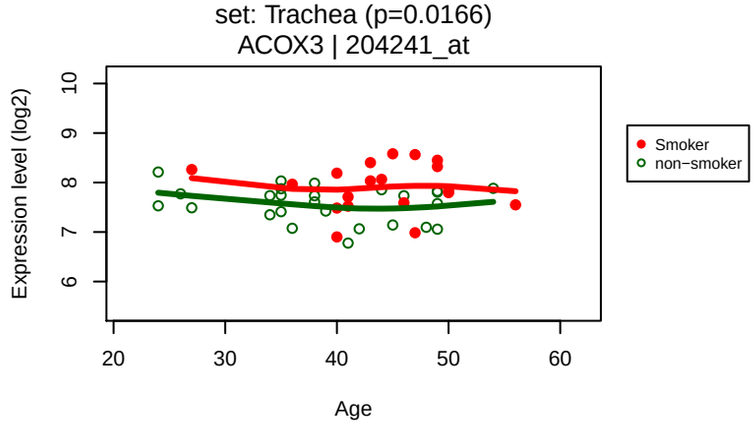

Gene expression ratio in Males
gene: ACOX3 | 204241_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0322
p=0.6722
p=0.0483
p=0.3202
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ACOX3 | 204241_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5764
p=0.1197
p=0.6021
p=0.0454
SmallAirway2
LargeAirway2
Trachea
Macrophage2

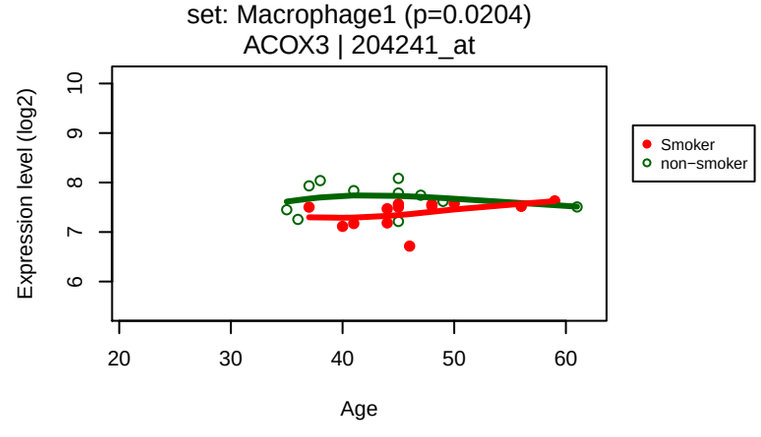

set: SmallAirway1 (p=0.3248)
ACOX3 | 204241_at
Expression level (log2)
Age
Smoker
non−smoker

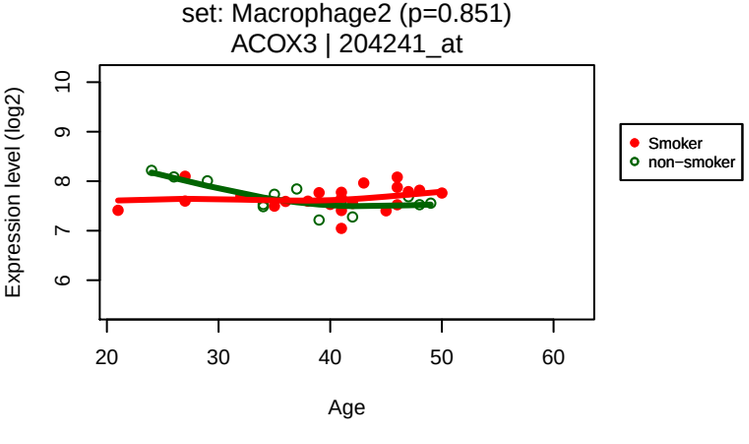

set: SmallAirway2 (p=0.2287)
ACOX3 | 204241_at
Expression level (log2)
Age
Smoker
non−smoker

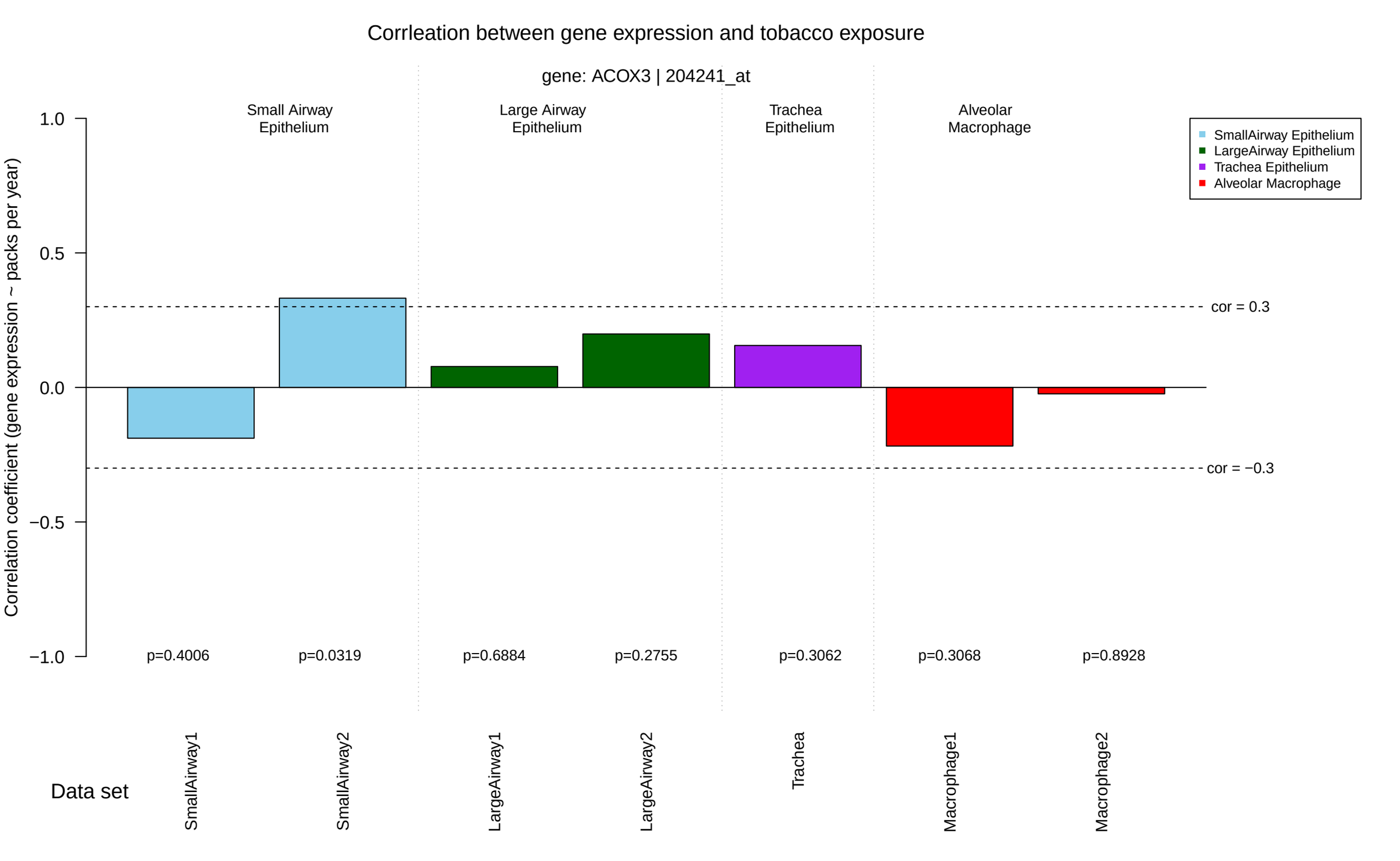

set: LargeAirway1 (p=0.2941)
ACOX3 | 204241_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2906)
ACOX3 | 204241_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0166)
ACOX3 | 204241_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0204)
ACOX3 | 204241_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.851)
ACOX3 | 204241_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ACOX3 | 204241_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4006
p=0.0319
p=0.6884
p=0.2755
p=0.3062
p=0.3068
p=0.8928
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2